

Solving Equations with Square Roots (Homework)

Solve.

Answers

① $3 = 1 + \sqrt{x}$

$x = 4$

② $5\sqrt{y} = 40$

$y = 64$

③ $8 = 2\sqrt{b} - 6$

$b = 49$

④ $\frac{\sqrt{x}}{3} + 5 = 2$

No Solution

⑤ $6 - 9\sqrt{y+1} = -12$

$y = 3$

⑥ $7 = 1 + \sqrt{3b}$

$b = 12$

⑦ $3\sqrt{2x-1} + 4 = 9$

$x = 17\frac{1}{9}$

⑧ $7 = 6 + \sqrt{2 + \frac{y}{5}}$

$y = -5$

⑨ $\frac{\sqrt{3b+2}}{4} - 1 = -3$

No Solution

⑩ $\frac{\sqrt{x-5}}{-2} + 9 = 7$

$x = 21$

⑪ $\sqrt{y^2 - 17y + 72} = 0$

$y = 8 \text{ or } 9$

⑫ $3 = \sqrt{b^2 + 7b + 1}$

$b = 1 \text{ or } -8$

⑬ $3\sqrt{2-4x} + 2 = -1$

No Solution

⑭ $\frac{\sqrt{\frac{y}{2} + 3}}{6} + 7 = 8$

$y = 66$

⑮ $x - 2 = \sqrt{-7x + 14}$

$x = 2, x \neq -5$

⑯ $\sqrt{y+3} - 3 = y$

$y = -2 \text{ or } -3$

⑰ $\sqrt{2b+8} = \sqrt{3b}$

$b = 8$

⑱ $\sqrt{2b} = \sqrt{4b+2}$

No Solution

⑲ $\sqrt{x^2 + 18} = \sqrt{11x}$

$x = 9 \text{ or } 2$

⑳ $\sqrt{y+3} = 4\sqrt{y}$

$y = \frac{1}{5}$

㉑ $2\sqrt{8b+6} = 5\sqrt{b-1}$

No Solution

㉒ $\frac{\sqrt{2x+7}}{3} = \sqrt{x}$

$x = 1$

㉓ $-2\sqrt{y} + 4 = -12$

$y = 64$

㉔ $\sqrt{b+6} = \sqrt{b} + 3$

No Solution

㉕ $\sqrt{3x+6} = \sqrt{3x} + 2$

$x = \frac{1}{12}$

㉖ $\sqrt{5y-2} - 1 = \sqrt{-7+5y}$

$y = \frac{11}{5}$